Vinith Kuruppu

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Education

University of California, Berkeley - Master of Science in Data Science (GPA: 3.97/4.0) Dec 2025
- **Honors:** Hal R. Varian Award – Distinguished Capstone Project
- **Relevant Coursework:** Machine Learning, Natural Language Processing, Computer Vision, Generative AI, Research Design and Application, Data Engineering, Experiments and Causal Inference

Loyola University Chicago - Bachelor of Science in Computational Neuroscience May 2022

Skills & Technologies

Languages: Python, R, SQL, Bash
Machine Learning & AI: PyTorch, TensorFlow, Hugging Face Transformers, scikit-learn, XGBoost, QLoRA/LoRA (PEFT), RAG (dense/sparse/hybrid retrieval, cross-encoder reranking), FAISS, Vision Transformers, CNNs, ASR, Large Language Models (LLMs)
Research & Statistics: Statistical Modeling, Ablation Studies, Experimental Design, A/B Testing, Causal Inference, Regression Modeling, Hypothesis Testing, Effect Sizes & Confidence Intervals, Power Analysis, Fairness Analysis & Bias Mitigation
Systems & Data: AWS (SageMaker, EC2, S3), Spark/PySpark, Docker, Git, PostgreSQL, MongoDB, MLflow, Weights & Biases, NumPy, Pandas

Experience

Machine Learning Researcher, University of California, Berkeley | Remote Aug 2025 - Present
- Built a pediatric clinical decision-support system that retrieves, reranks, and synthesizes evidence from medical literature to generate grounded answers for physician diagnostic queries.
- Designed a reproducible evaluation framework spanning retrieval recall, clinical relevance, and answer grounding; benchmarked 54 pipeline configurations with versioned experiment tracking to enable systematic comparisons.
- Selected the final pipeline architecture by quantifying faithfulness–completeness tradeoffs across retrieval strategies, rerankers, and generator LLMs, balancing answer quality against runtime constraints.

AI/ML Research Associate, RAND Corporation | Remote May 2024 - May 2025
- Developed fairness-aware allocation models that informed the allocation of $70 billion across 3,300+ hospitals, reducing allocation error and improving equity in disaster reimbursement across U.S. regions.
- Shipped an Excel-integrated classification tool backed by an AWS ML endpoint, enabling non-technical analysts to label 1M+ expense records with one click; improved accuracy by 22% vs. manual baseline and was adopted in FEMA allocation review workflows.
- Designed simulation frameworks for disaster-response scenarios under uncertainty and ran sensitivity analyses across 50+ parameter configurations; findings directly shaped FEMA preparedness protocols.

Research Data Scientist, Exponent Incorporated | Chicago, IL Jul 2022 - May 2024
- Engineered data ingestion pipelines processing high-frequency physiological signals from 3,000+ Apple Watch devices, reducing integration time by 15% and enabling rapid iteration on Apple Health's hypertension detection algorithms.
- Led end-to-end FDA-compliant validation for biosensor clinical trials, achieving 100% pass rate on QA audits through rigorous statistical validation and bias analysis; work enabled product launch for a health monitoring platform.
- Conducted fairness-aware analysis of biosensor signals across demographic subgroups (age, gender, skin tone), identifying algorithm performance disparities that informed design changes, improving signal-quality acceptance by 25% and reducing group-level performance gaps.

Computer Vision Research Assistant, Loyola University Chicago | Chicago, IL Jan 2020 - Jun 2022

- Led a 6-person research team developing bio-inspired CNN architectures with attention mechanisms to model human visual perception, establishing lab benchmarks for human behavioral data from EEG/eye-tracking studies.
- Improved model accuracy by 11% vs. prior lab baseline via targeted augmentation, systematic hyperparameter optimization, and architecture refinements on object recognition tasks.
- Engineered Python data pipelines for image and behavioral datasets, reducing preprocessing time by 30% through parallelization and automated quality checks.

Projects

SurgiRAG: Retrieval-Augmented Generation for Surgical Question Answering 2025

PyTorch, LLaMA-3.2-11B (QLoRA), BioBERT, BGE, FAISS, Hugging Face

- LoRA-fine-tuned LLaMA-3.2-11B on surgical literature and built a domain-adaptive RAG pipeline (BioBERT retrieval + cross-encoder reranking), improving LLM-judge faithfulness from 0.25 to 0.66 vs. the base model.
- Designed an LLM-as-judge evaluation framework (GPT-4o-mini) to assess faithfulness vs. fluency tradeoffs not captured by ROUGE/BLEU.
- Conducted a 12-variant ablation study across retrieval, reranking, and generation modules on a curated 32-item benchmark with a 1,527-chunk surgical corpus, identifying cross-encoder reranking as the primary driver of faithfulness gains.

AgeVoicE: Adapting Whisper for Disfluent Speech (Hal R. Varian Award Winner) 2025

PyTorch, Whisper, LoRA, Hugging Face, AWS

- Fine-tuned Whisper ASR models using LoRA for older-adult and dementia-affected speech, improving WER by 17% vs. baseline; deployed via Hugging Face Inference API and an interactive web demo for real-time transcription.
- Built an end-to-end AWS training pipeline with distributed training, custom preprocessing (noise removal, disfluency handling, audio normalization), experiment tracking, and systematic hyperparameter sweeps (LR, LoRA rank, dropout).
- Performed stratified failure analysis across acoustic conditions and speaker characteristics using regression modeling and hypothesis testing to identify drivers of elevated WER; findings guided targeted data augmentation to mitigate demographic bias.

PathoVision: Hybrid Vision Transformer for Brain Tumor Classification 2025

PyTorch, DINOv2 (Vision Transformer), scikit-learn, OpenCV

- Achieved 96.7% test accuracy on multiclass brain tumor MRI classification using a hybrid pipeline combining DINOv2 embeddings and handcrafted features (Canny edges, DoG) with lightweight classifiers, outperforming CNN baselines in the low-label regime.
- Benchmarked 40+ feature–classifier combinations; showed that logistic regression on ViT embeddings matches or exceeds CNN performance while training in seconds, demonstrating a low-compute alternative for small-data settings.
- Conducted error analysis across tumor subtypes and imaging sources, identifying contrast and orientation artifacts as primary misclassification drivers and guiding preprocessing decisions that improved cross-source accuracy.